Following our meeting on 11/5/2026, I suggest the following:

Stage 1: Replace the heavy DLoc encoder-decoder (~15M params) with an edge-friendly lightweight architecture.

You should test the MobileNetV2 Localization Network:

Suggested Architecture:

Encoder:

- MobileNetV2 backbone
- input channels modified: Conv2d(4, 32, kernel_size=3)
- Decoder:

  Upsample

  Conv(320 → 128)

  Upsample

  Conv(128 → 64)

  Upsample

  Conv(64 → 32)

  Conv(32 → 1)

  Sigmoid

Stage 2 Distillation

Goal

Transfer localization knowledge from DLoc teacher → lightweight student.

Distillation Setup

Teacher: Your corrected DLoc baseline.

Student: MobileNetV2 localization model.

Use combined supervision:
L=α*L_GT+β*L_KD

Where:

Ground Truth Loss: MSE(student_output, ground_truth)

Distillation Loss: MSE(student_output, teacher_output) start with $\alpha = 0.7$, $\beta = 0.3$

Stage 3 Efficiency Evaluation

Goal

Convert the work into a systems-aware conference paper.

You MUST compare:

Metric	Why Important
Parameters	model size
FLOPs	compute cost
Inference latency	real-time deployment
GPU memory	edge feasibility
Localization error	performance

Also you can replace the MobileNetV2 with our own Tiny CNN with the following architecture then follow the same distillation steps but this step is optional

Stage	Layer / Operation	Output Shape	Purpose
Input	4-channel AoA heatmaps	4×64×644 \times 64 \times 644×64×64	CSI/AoA spatial input from 4 APs
Block 1	Conv2D (4→16, 3×3) + ReLU	16×64×6416 \times 64 \times 6416×64×64	Initial feature extraction
	Depthwise Separable Conv (16→16) + ReLU	16×64×6416 \times 64 \times 6416×64×64	Efficient spatial filtering
	MaxPool2D (2×2)	16×32×3216 \times 32 \times 3216×32×32	Downsampling
Block 2	Depthwise Separable Conv (16→32) + ReLU	32×32×3232 \times 32 \times 3232×32×32	Higher-level feature learning
	Depthwise Separable Conv (32→32) + ReLU	32×32×3232 \times 32 \times 3232×32×32	Channel refinement
	MaxPool2D (2×2)	32×16×1632 \times 16 \times 1632×16×16	Downsampling
Block 3	Depthwise Separable Conv (32→64) + ReLU	64×16×1664 \times 16 \times 1664×16×16	Compact latent representation
Decoder 1	ConvTranspose2D (64→32, stride=2) + ReLU	32×32×3232 \times 32 \times 3232×32×32	Upsampling
Decoder 2	ConvTranspose2D (32→16, stride=2) + ReLU	16×64×6416 \times 64 \times 6416×64×64	Spatial reconstruction
Output Head	Conv2D (16→1, 1×1) + Sigmoid	1×64×641 \times 64 \times 641×64×64	Predicted localization heatmap

MAMBA implementation:

In a Mamba-enhanced version of DLoc, the **Mamba module should replace the deep ResNet bottleneck portion of the encoder-decoder pipeline**, not the entire network.

- CNN handles **local spatial structure**
- Mamba handles **global signal relationships**
- Decoder reconstructs **precise heatmap localization**

The key idea is:

- keep the efficient low-level spatial feature extraction of CNNs,
- replace the heavy/global context modeling part with Mamba,
- preserve the heatmap prediction framework.

Current DLOC pipeline is as follow:

```
AoA Heatmaps (4×161×361)
        ↓
Initial CNN Layers
        ↓
ResNet Encoder
        ↓
6 Residual Blocks
        ↓
Latent Feature Map
        ↓
Decoder
        ↓
Localization Heatmap
```

We will replace 6 ResNet Bottleneck Blocks with: Mamba Sequence Modeling Blocks

Decoder remains CNN-based.

The final pipeline is as follow

Component	Best Choice
Low-level spatial extraction	CNN
Global spatial reasoning	Mamba
Heatmap generation	CNN decoder

Stage	Component	Operation	Output Shape	Purpose
Input	CSI / AoA Heatmaps	Raw WiFi feature maps (4 AP channels)	[B, 4, 161, 361]	Input signal representation
Stem Encoder	Conv Block 1	Conv(4→32, 3×3) + BN + ReLU	[B, 32, 161, 361]	Low-level feature extraction
	Conv Block 2	Conv(32→64, 3×3, stride=2) + BN + ReLU	[B, 64, 81, 181]	Downsampling + feature compression
	Conv Block 3	Conv(64→128, 3×3, stride=2) + BN + ReLU	[B, 128, 41, 91]	Compact spatial embedding

Stage	Component	Operation	Output Shape	Purpose
Reshape	Tokenization	Flatten spatial dims → sequence	[B, 3731, 128]	Convert image → sequence for global modeling
Global Modeling (REPLACEMENT CORE)	Mamba Block × 4	State-space sequence modeling (selective scan)	[B, 3731, 128]	Global multipath + NLOS reasoning
Reshape Back	Spatial Reconstruction	Reshape tokens back to feature map	[B, 128, 41, 91]	Restore spatial structure
Decoder Stage 1	Upsample + Conv	Conv(128→64, 3×3)	[B, 64, 81, 181]	Spatial refinement
Decoder Stage 2	Upsample + Conv	Conv(64→32, 3×3)	[B, 32, 161, 361]	High-resolution reconstruction
Output Layer	Heatmap Head	Conv(32→1, 1×1) + Sigmoid	[B, 1, 161, 361]	Final localization heatmap